

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electronics and Computer Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Electronics and Computer Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTECPE502 - DIGITAL SIGNAL & IMAGE PROCESSING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is a correct definition of a signal?
A. A system that processes data
B. A device that transforms energy
C. A function representing a physical quantity varying with time or space
D. A waveform used only in audio processing
A signal is even if:
A. $x(t) = -x(t)$ B. $x(t) = x(-t)$
C. $x(t) = 0$ D. $x(t) = x(t+1)$
- 3 A signal is said to be periodic if:
A. $x(t) = x(-t)$ B. $x(t) = 0$
C. $x(t+T) = x(t)$ for some period T D. It is constant
- 4 The impulse signal $\delta(t)$ is also known as:
A. Sine wave B. Unit step
C. Dirac function D. Ramp function
- 5 Time shifting of $x(t) = x(t - t_0)$ causes:
A. Time compression B. Frequency doubling
C. Shift by t_0 units D. Signal to become zero
- 6 Nyquist theorem states that sampling rate must be:
A. Equal to signal frequency
B. Half of signal frequency
C. Twice the maximum frequency present in signal
D. One-fourth of the bandwidth
- 7 A system is causal if:
A. Output depends on future input
B. Output depends only on current and past inputs
C. Output is a constant
D. It processes no input

- 8 The DTFT of a signal is:
 A. Always periodic with period 1
 B. Always periodic with period 2π
 C. A finite-length sequence
 D. Non-periodic
- 9 Which transform is suitable for implementing FFT algorithms?
 A. DTFT
 B. Laplace
 C. DFT
 D. Z-transform
- 10 Which type of image contains only two possible intensity levels?
 A. RGB image
 B. Grayscale image
 C. Binary image
 D. Color image
- 11 The Z-transform is defined for signals that are:
 A. Continuous-time only
 B. Discrete-time only
 C. Random
 D. Noise signals
- 12 The primary goal of digital image processing is to:
 A. Compress images
 B. Convert analog to digital
 C. Improve the visual appearance of images or prepare them for analysis
 D. Change image formats
- Q2. Solve the following.
- A) a. Given Signal is $x(t)=\sin(200\pi t)$, find the minimum sampling rate.
 b. Check whether $y(t)=3x(t)+2$ is Linear or Non-linear. 6
- B) Find the energy of the signal $x(t)=e^{-2t}u(t)$. 6
- Q3. Solve the following.
- A) Compute 4-point DFT of $x[n]=\{1,2,3,4\}$. 6
- B) Find the Z-transform and ROC of $x[n]=a^n u[n]$, $|a| < 1$. 6
- Q4. Solve Any Two of the following.
- A) Explain the fundamental steps involved in digital image processing. 6
- B) Define the following pixel relationships with examples: 6
- (i) Adjacency
 (ii) Connectivity
 (iii) Neighborhood
- C) Explain and compare the following distance metrics and also find them for $p=(2,3), q=(5,1)$. 6
- (i) Euclidean Distance
 (ii) City-block Distance
 (iii) Chessboard Distance
- Q5. Solve Any Two of the following.
- A) Explain the following intensity transformations with graphs: 6
- (a) Image Negative
 (b) Log Transformation
 (c) Power-law Transformation
- B) Compare and explain box filter, Gaussian filter, and median filter. 6
- C) Define 1D and 2D DCT. Why is DCT preferred in image compression? 6
- Q6. Solve Any Two of the following.

- A) Explain the principles of point, line, and edge detection. Give suitable masks for each. 6
- B) Explain local and global thresholding. Apply threshold $T=128$ to image vector: $\{100,130,90,180,200,50\}$ 6
- C) Explain segmentation using region splitting and merging with an example. 6

*** End ***